ABSTRACT

The spin-orbit coupling has long been known to significantly impact the properties of solid-state systems. Recent developments and proposals in the field of quantum technology seek to exploit spin-orbit coupling to further advance solid-state quantum devices. Still, for us to effectively harness the potential of the spin-orbit coupling, the effect must be studied further. In the present thesis, we study the spin-orbit coupling of p-like hole states in a uniaxially strained germanium heterostructure with an externally applied constant electric field directed along the strain axis. Specifically, we derive, using a second-order Schrieffer-Wolff transformation, an explicit expression for the strength of the effective spin-orbit coupling in the heavy-hole low-energy subspace of the system. The spin-orbit coupling is induced by the electric field. It is thus tunable, making such a system attractive for applications such as semiconductor spin quantum bits or superconducting hybrid structures. By comparing the analytical expression we derive with numerical results, we find that our expression captures the maximum magnitude of the spin-orbit coupling for small electric fields and a narrow confinement in the direction of the electric field, and qualitatively matches its dependence on the size of the heterostructure, the electric field strength, and the strain. Our approach thus gives an overall insight into the behavior of the spin-orbit effect in a low-energy semiconductor environment.

SAMMENDRAG

Det har lenge vært kjent at spinn-banekoplingen har en betydelig påvirkning på egenskapene til faststoffsystemer. Nylige utviklinger innen kvanteteknologi forsøker å utnytte spinn-banekoplingen for å videreutvikle faststoff-kvanteenheter. Likevel må effekten studeres videre for å utnytte potensialet til spinn-banekoplingen effektivt. I denne oppgaven studerer vi spinn-banekoplingen til hulltilstander med totalt dreieimpulsmoment lik 3/2 i en germanium-heterostruktur med enakset tøyning og et eksternt, konstant elektrisk felt i tøyningsretningen. Vi utleder spesifikt, ved hjelp av en andreordens Schrieffer-Wolff-transformasjon, et eksplisitt uttrykk for styrken til den effektive spinn-banekoplingen i lavenergi-underrommet for tunge hull. Spinn-banekoplingen induseres av det eksterne elektriske feltet og er derfor justerbar, noe som gjør systemet vi studerer attraktivt for anvendelser, for eksempel innen halvlederspinn-kvantebits eller superledende hybridstrukturer. Ved å sammenligne vårt analytiske uttrykk med numeriske utregninger finner vi at uttrykket vi har utledet fanger spinn-banekoplingens maksimale størrelse når det elektriske feltet er svakt og innesperringen i det elektriske feltets retning er smal. Videre samsvarer det uttrykket vi har utledet kvalitativt med spinn-banekoplingens avhengighet av heterostrukturens størrelse, styrken til det eksterne elektriske feltet og tøyningen. Tilnærmingen vår gir dermed en overordnet innsikt i oppførselen til spinn-banekoplingen i lavenergibeskrivelsen av et halvledersystem.

PREFACE

I want to express my deepest gratitude to my supervisor, Prof. Jeroen Danon, for supervising me during the writing of my master's thesis. It has been a tremendous journey, and his help and guidance has been invaluable.

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With these keystrokes, I conclude six years of university, and I can say, with my whole heart, that should I be brought back in time six years from now, I would do it again.

"If you think you understand quantum mechanics,
you don't understand quantum mechanics." —

Richard P. Feynman

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **1D** One-dimensional
- **2D** Two-dimensional
- **ESC** Einstein summation convention
- **Ge** Germanium
- **HH** Heavy hole
- **LH** Light hole
- **LKBP** Luttinger-Kohn-Bir-Pikus
- **PIAB** Particle-in-a-box
- **QDPT** Quasi-degenerate perturbation theory
- **Qubit** Quantum bit
- **SC** Superconductor
- **SE** Schrödinger equation
- **SO** Spin-orbit
- **SOC** Spin-orbit coupling
- **SOV** Spin-orbit velocity
- **SW** Schrieffer-Wolff
- **SWT** Schrieffer-Wolff transformation
- **TAM** Total angular momentum

CHAPTER

ONE

INTRODUCTION

It was L. H. Thomas who first derived a spin-orbit energy splitting of the electron spectrum in a hydrogen atom that matched experimental results and the predictions of the Dirac equation [1, 2]. The spin-orbit coupling (SOC), also referred to as the spin-orbit interaction, arises due to the coupling between a spinful particle's motion through an electric field. It is a relativistic effect and can be understood as the spin coupling to the magnetic field being induced by the electric field because *it* is moving in the particle's rest frame. What Thomas showed was that relativistic considerations are needed for accurate predictions of the effect, even though a classical model can predict the existence and form of the effect [3]. Inversion asymmetry of the system breaks the degeneracy of angular momentum states and makes the energy spectrum dependent on spin and total angular momentum [4, 5]. Thus, a spin-dependent splitting can be induced without a magnetic field present. This effect is often on the same order as other primary effects in a system, and it can thus alter the behavior and properties of solids in significant and interesting ways. In systems of reduced dimensions, the effect is amplified even more due to the introduced breaking of inversion symmetry at and near interfaces [6]. Due to the significant effect the SOC can have on materials, it has become a topic of great interest for, for example, spintronics and condensed matter physics.

There are several ways of breaking inversion symmetry in a solid-state system, giving rise to different kinds of SOC. If the structure breaks inversion symmetry of the bulk material, we call this bulk inversion asymmetry. One example of SOC arising from this asymmetry is the Dresselhaus SOC [7], observed, for example, in materials with a zinc blende crystal structure, such as GaAs. The zinc blende structure is not centrosymmetric and thus breaks inversion symmetry. Furthermore, if inversion symmetry is broken by structural factors such as interfaces or external fields, known as structure inversion asymmetry, effects such as the Bychkov-Rashba SOC arise [8]. This type of SOC can also be present in materials that are bulk inversion symmetric, for example near surfaces and interfaces. Both of these effects have been thoroughly investigated, but new proposals and advancements in condensed matter physics and its applications may require an understanding of higher-order spin-orbit effects [9]. This exemplifies the need for further research into the effects of the SOC.

In normal-state materials such as semiconductors and metals, the SOC is the source of a whole host of phenomena. In three dimensions, effects such as the family of spin Hall effects [10, 11] are prominent and have already shown great utility in technologies such as spin-orbit torque MRAM [12]. Furthermore, effects such as spin-charge conversion [13] and topological insulating states [14] are proving useful in the development of new technologies. In systems of reduced dimensions, spin-orbit effects are, as mentioned, amplified and give rise to new effects and applications of the SOC, such as Edelstein effects [15] and non-collinear spin textures such as skyrmions and chiral domain walls [6]. The potential of the SOC is also being used in quantum computation, in hole-spin semiconductor quantum bits (qubits). A qubit is a quantum two-level system, meaning a physical system in the quantum regime with two orthogonal eigenstates. In the pursuit of making solid-state qubits, two main approaches are being explored: superconductor (SC) and semiconductor qubits. These two approaches differ in the medium in which the two-level system constituting a qubit is realized, namely, a superconductor or a semiconductor. Within the class of semiconductor qubits, we have the so-called hole-spin semiconductor qubits. These are qubits realized in semiconductors, where the qubit states are encoded in the eigenstates of holes in the valence band. By applying an electric field to such a system, one can induce strong SOC, which can then be manipulated and thus controlled by altering the system's electric confinement. Since the SOC can alter the spin and orbital angular momentum of the hole, we now have a way of manipulating the state of the qubit using all-electrical control. This could be beneficial, as alternative ways of controlling a qubit, such as magnetic fields, can be cumbersome to operate and control. The downside of relying on the SOC, however, is that its behavior is more complex and thus harder to predict.

In addition to normal-state materials, the SOC is the source of several exotic phenomena in SCs and hybrid SC systems, with *hybrid* referring to the combination of SCs with other states of matter, such as ferromagnets. The SOC in SCs can, for example, be observed to enable various magnetoelectric effects [16] and to enhance spin Hall effects. In hybrid structures, the SOC is a critical component in inducing triplet superconductivity, without, for example, having to manufacture complex non-collinear magnetic structures at the interface between a SC and a ferromagnet. The phenomenon is central to the quest for dissipationless spin currents [17], topological superconductivity [18], and the elusive Majorana states, which are predicted to unlock a new frontier of fault-tolerant topological quantum computing [19]. In Josephson junctions a strong SOC is also proving itself useful for applications such as non-binary random access memory [20] and low-power high speed superconducting logic and quantum computation [21, 22].

From the examples above, it is apparent that effectively manipulating the SOC in solid-state devices can open the door to a plethora of new opportunities and applications in spintronics and quantum technology. These fields, however, are not yet mature enough for widespread commercial use [23]. This is why the work presented in this thesis aims to further our understanding of the SOC in solid-state systems. We do this by deriving an explicit perturbative expression for the spin-orbit strength, or spin-orbit velocity (SOV) v, of the degenerate ground state

of a valence band hole state in a strained 2-dimensional (2D) semiconductor with zinc-blende or diamond structure; a face-centered cubic lattice with two atoms in its basis. This is done using a Schrieffer-Wolff transformation (SWT) on the isotropic 4×4 Luttinger-Kohn-Bir-Pikus (LKBP) Hamiltonian for a p-like hole state with a total angular momentum (TAM) of 3/2, in addition to a rectangular abrupt confining potential V_c in the xz-plane and a constant electric field E. This Hamiltonian is written as

$$H_{\text{LKBP}} = \left(\gamma_1 + \frac{5\gamma_s}{2}\right)\frac{\boldsymbol{p}^2}{2m} - \frac{\gamma_s}{m}\big(\boldsymbol{p}\cdot\boldsymbol{J}\big)^2 + |b|\varepsilon_0 J_z^2 - eEz + V_c \tag{1.1}$$

where m and e are the electron mass and charge, γ_1 and γ_s are the material parameters of the system, $\boldsymbol{p}$ is the canonical momentum, $\boldsymbol{J}$ is the vector of TAM 3/2 matrices, $|b|$ and ε_0 are strain parameters, and E is a constant electric field along z. There exists a more general form of Equation 1.1 where there are three material parameters γ_1, γ_2, and γ_3 present. In Equation 1.1 we employ the spherical approximation where $\gamma_2 = \gamma_3 \equiv \gamma_s$. This only works for materials where γ_2 and γ_3 are similar enough so that such an approximation is justifiable. An example of a material where the spherical approximation may be used is Ge, which is precisely the material we aim to investigate in this thesis.

Further, we use quasi-degenerate perturbation theory to derive an effective one-dimensional (1D) Hamiltonian for the two lowest states in the free y-direction in the form

$$H_W = \frac{p_y^2}{2m^*} + p_y \boldsymbol{v}\cdot\boldsymbol{\sigma}, \tag{1.2}$$

where m^* is the effective mass of the states, $\boldsymbol{\sigma}$ is the vector of Pauli spin matrices, and $\boldsymbol{v}$ is the aforementioned SOV that describes the spin and alignment of the SOC of the system. This work is heavily inspired by the works of Bosco et al. [24] and Bosco, Hetenyi, and Loss [25].

Furthermore, we compute numerical values of the SOV to compare and verify our analytical results. This is done by discretizing Equation 1.1 and projecting the terms that are linear or quadratic in p_y onto the two lowest energy eigenstates of the part that is independent of p_y.

We will start by presenting some relevant theory for the work we have done in chapter 2. This includes sections on the SOC, quasi-degenerate perturbation theory via the Schrieffer-Wolff transformation (SWT), and an outline of the derivation of the Luttinger-Kohn-Bir-Pikus Hamiltonian. We then describe in chapter 3 the methods we have employed to obtain our results. In this chapter, we include a more rigorous formulation of the problem we aim to solve, a description of the numerical methods we use, and an outline of the method we use to derive perturbative results. In chapter 4 we present the analytic results we have derived and compare them to numerical results. In chapter 5 we discuss central assumptions and constraints of our model and results, and finish by presenting some further work that may be done in section 5.4. We relate in the latter chapter our theory

to two specific applications: a strained germanium hole spin qubit related to the one presented in [24] and a hybrid SC qubit with inspiration from [26].

1.1 Conventions

- We omit hat notation on operators and denote matrices with underlines. We would, for example, denote the Hamiltonian operator by H and the same operator in matrix form as $\underline{H}$
- We denote the z-component of the TAM of a state by m_j or μ_j, inferring that $m_j, \mu_j = \pm\frac{3}{2}, \pm\frac{1}{2}$
- When we want to reference an element of a matrix, we will enclose it in parentheses and use subscripts for indices. We will differentiate between the indices using a comma. In some cases, however, we want to index using the quantum numbers an index refers to. We will then differentiate between *quantum numbers* using a comma and between *indices* using a semicolon. We could, for example, refer to the element of a Hamiltonian as $(\underline{H})_{i,j}$ or as $(\underline{H})_{n_x,n_z,m_j;\nu_x,\nu_z,\mu_j}$
- We use α's to refer to the letters x, y, and z. Additionally, we let $r_\alpha = x, y, z$ refer to the spatial directions and $p_\alpha = p_x, p_y, p_z$ refer to the canonical momentum operators.
- We utilize the Einstein summation convention in some expressions, meaning we write, for example, $n_\alpha^2 \varepsilon_\alpha$ in place of $\sum_{\alpha\in\{x,y,z\}} n_\alpha^2 \varepsilon_\alpha$
- In expressions in operator form, we omit writing out tensor products and let them be inferred from the expressions. For example, p_x^2 is, in Equation 1.1, actually $p_x \otimes \mathbf{1}_{4\times4}$. On the same note, in matrix form we do write out tensor products. In these cases $\otimes$ is a Kronecker product
- In this thesis we use the TAM-3/2 matrices. During our numerical calculations we use the conventional ones where the basis is in the order $\{|3/2, 3/2\rangle, |3/2, 1/2\rangle, |3/2, -1/2\rangle$ whilst in our analytical derivation we reorder the basis to be $\{|3/2, 3/2\rangle, |3/2, -3/2\rangle, |3/2, 1/2\rangle, |3/$ The matrices we thus use in our analytical calculations are

$$J_x = \frac{1}{2}\begin{pmatrix} 0 & 0 & \sqrt{3} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ \sqrt{3} & 0 & 0 & 2 \\ 0 & \sqrt{3} & 2 & 0 \end{pmatrix}, \tag{1.3a}$$

$$J_y = \frac{i}{2}\begin{pmatrix} 0 & 0 & -\sqrt{3} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ \sqrt{3} & 0 & 0 & -2 \\ 0 & -\sqrt{3} & 2 & 0 \end{pmatrix}, \tag{1.3b}$$

$$J_z = \frac{1}{2}\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{1.3c}$$

CHAPTER

TWO

THEORY

As background for the work presented in this thesis, we show how to derive the SOC term of the SE from the non-relativistic limit of the Dirac equation. We then use this expression to outline the derivation of the LK Hamiltonian, which describes holes in the valence band of a zinc-blende semiconductor. Finally, we describe how strain is introduced into the Hamiltonian to arrive at the LKBP Hamiltonian, which is the Hamiltonian we will study in our work.

2.1 Spin-Orbit Interaction

The SOC is a relativistic effect. A non-relativistic term for the SO of a spin-1/2 particle may thus be derived from the Dirac equation in the non-relativistic limit. In the present section, we show how to derive the SO term in the non-relativistic SE. This is done as background for deriving the Hamiltonian that describes SO in the valence band of a semiconductor.

The Dirac equation in an electric field with scalar potential V becomes [17]

$$\begin{pmatrix} mc^2 + V & -i\hbar c\boldsymbol{\sigma} \cdot \nabla \\ -i\hbar c\boldsymbol{\sigma} \cdot \nabla & mc^2 + V \end{pmatrix} \begin{pmatrix} \psi_e \\ \psi_h \end{pmatrix} = (\varepsilon + mc^2) \begin{pmatrix} \psi_e \\ \psi_h \end{pmatrix}. \tag{2.1}$$

To obtain the classical Hamiltonian including SOC, we must take the second-order approximation in the classical limit where $\varepsilon, V \ll mc^2$. We write out Equation 2.1:

$$(V - \varepsilon)\psi_e - i\hbar c\boldsymbol{\sigma} \cdot \nabla\psi_h = 0, \tag{2.2a}$$

$$(V - \varepsilon - 2mc^2)\psi_h - i\hbar c\boldsymbol{\sigma} \cdot \nabla\psi_e = 0. \tag{2.2b}$$

Then we take Equation 2.2b, solve it for ψ_h and Taylor expand the resulting expression to get

$$\psi_h \approx -\frac{i\hbar}{4m^2c^2}\left(1 + \frac{V - \varepsilon}{2mc^2}\right)\boldsymbol{\sigma} \cdot \nabla\psi_e. \tag{2.3}$$

By substituting Equation 2.3 into Equation 2.2a and using the Pauli identity

$$(\boldsymbol{a}\cdot\boldsymbol{\sigma})(\boldsymbol{b}\cdot\boldsymbol{\sigma}) = (\boldsymbol{a}\cdot\boldsymbol{b})\mathbb{1} + i(\boldsymbol{a}\times\boldsymbol{b})\cdot\boldsymbol{\sigma}, \tag{2.4}$$

the product rule, and by discarding all other terms, we get the Schrödinger equation with SO for the electron wavefunction,

$$\left[\frac{p^2}{2m} + V(r) + \frac{\hbar}{4m^2c^2}\boldsymbol{\sigma}\cdot(\nabla V\times\boldsymbol{p})\right]\psi_e = \varepsilon\psi_e, \tag{2.5}$$

and thus the Hamiltonian

$$H_{\mathrm{SO}} = \frac{\hbar}{4m^2c^2}\boldsymbol{\sigma}\cdot(\nabla V\times\boldsymbol{p}). \tag{2.6}$$

This is the lowest-order SO term and is linear in p. Based on the nature of V, different kinds of SO arise. If the microscopic potential is inversion asymmetric, the Dresselhaus SOC arises. If the macroscopic potential or some other external electric field breaks inversion symmetry, the Rashba SO effect arises. This derivation of the Pauli-SOC is given as background, as Equation 2.6 will be further used to outline the derivation of the SOC in semiconductors in section 2.3.

2.2 Quasi-Degenerate Perturbation Theory

An indispensable tool in a physicist's toolbelt is perturbation theory, as it lets one treat complex systems as long as one can split the system's Hamiltonian into an exactly solvable part and a small perturbation. For the problem we want to investigate, in which we aim to derive an effective Hamiltonian for a small subset of our system's eigenbasis, quasi-degenerate perturbation theory (QDPT) is required. What QDPT does is integrate out high-energy interactions from a low-energy subsystem, thus yielding an effective low-energy description of the subsystem. There are several QDPTs, but we shall use the SWT in our thesis due to its usefulness with Hamiltonians. The transformation is unitary and thus preserves the eigenvalues of a Hermitian matrix. In this section we outline the derivation of the SWT presented in [4].

We start by considering a finite time-independent Hamiltonian H. This Hamiltonian can be split into the sum of two distinct Hamiltonians.

$$H = H_0 + H', \tag{2.7}$$

where H_0 is an exactly solvable system with eigenenergies E_n and eigenstates $|\psi_n\rangle$, while H' is a perturbation. We assume that we may partition our basis into two subsets A and B such that they are split in energy and interact weakly through H'. Our perturbation theory aims to block-diagonalize the full Hamiltonian H so that the subspaces A and B are decoupled, and we may treat the low-energy block that acts on subspace A as an effective Hamiltonian for that subspace. Let us denote eigenstates of subspace A by $|\psi_m\rangle$ and eigenstates of subspace B by $|\phi_l\rangle$.

The aim is to construct a unitary operator e^{-S} such that we get a transformed Hamiltonian operator

$$\tilde{H} = e^{-S} H e^{S}, \tag{2.8}$$

such that elements $\langle \psi_m | \, \tilde{H} \, | \phi_l \rangle$ are zero up to a desired order of S. Let us further subdivide our matrix H into the form

$$H = H_0 + H_1 + H_2, \tag{2.9}$$

where H_0 is diagonal, H_1 is block diagonal such that subspaces A and B do not interact, and H_2 is anti block-diagonal such that it contain *only* interactions between subspaces A and B. The goal of our unitary operator e^{-S} is thus to make H_2 block-diagonal while preserving the block-diagonal form of $H_0 + H_1$. Expand e^S as

$$e^S = \sum_{i=0}^{\infty} \frac{S^i}{i!}. \tag{2.10}$$

For e^{-S} to be unitary, S must be anti-Hermitian, i.e. that $S^\dagger = -S$. This statement can be shown to be true from

$$\begin{aligned} (e^{-S})^\dagger e^{-S} = \mathbb{1} &\Leftrightarrow \sum_{\substack{i,j=0 \\ i+j=k}}^{\infty} \frac{1}{i!j!} \big((-S)^i\big)^\dagger (-S)^j = \mathbf{0} \\ &\Leftrightarrow \sum_{k=1}^{\infty} (-1)^k \sum_{i=0}^{\infty} \binom{k}{i} (S^\dagger)^i S^{k-i} = \mathbf{0} \\ &\Leftrightarrow (S^\dagger + S)^k = \mathbf{0} \Leftrightarrow S^\dagger + S = \mathbf{0} \Leftrightarrow S^\dagger = -S. \end{aligned} \tag{2.11}$$

We note also that we may choose, without loss of generality, an S that is also anti-block-diagonal in the same way as H_2; no non-zero elements at places that do not couple subspaces A and B. This claim is rigorously shown to be valid in [27]. By substituting Equation 2.10 into Equation 2.8 we get

$$\tilde{H} = \sum_{j=0}^{\infty} \frac{1}{j!} [H, S]^{(j)} = \sum_{j=0}^{\infty} \frac{1}{j!} [H_0 + H_1, S]^{(j)} + \sum_{j=0}^{\infty} \frac{1}{j!} [H_2, S]^{(j)}, \tag{2.12}$$

where we have defined commutators

$$[A, B]^{(j)} = [\ldots [[A, B], B], \ldots B]. \tag{2.13}$$

Furthermore, we may categorize which terms in Equation 2.12 will lie on the block-diagonal of $\tilde{H}$ and which terms will lie on the anti-block-diagonal. Since S is anti-block-diagonal, multiplying it by a block-diagonal matrix an odd number

of times yields an anti-block-diagonal matrix, while multiplying it an even number of times yields a block-diagonal matrix. The converse is true when multiplying S by another anti-block-diagonal matrix; an odd number of times yields a block-diagonal matrix and an even number of times yields an anti-block-diagonal one. With this said, if we denote the block-diagonal and the block-diagonal part of $\tilde{H}$ by $\tilde{H}_d$ and $\tilde{H}_a$ respectively, we get

$$\tilde{H}_d = \sum_{j=0}^{\infty} \frac{1}{(2j)!}[H_0 + H_1, S]^{(2j)} + \sum_{j=0}^{\infty} \frac{1}{(2j+1)!}[H_2, S]^{(2j+1)} \tag{2.14a}$$

$$\tilde{H}_a = \sum_{j=0}^{\infty} \frac{1}{(2j+1)!}[H_0 + H_1, S]^{(2j+1)} + \sum_{j=0}^{\infty} \frac{1}{(2j)!}[H_2, S]^{(2j)} \tag{2.14b}$$

The end goal is, as stated before, to derive an S that makes $\tilde{H}$ block-diagonal. We thus require that $\tilde{H}_a = 0$. We assume we may expand S as

$$S = S^{(1)} + S^{(2)} + S^{(3)} + \ldots, \tag{2.15}$$

where the superscripts denote the order of each term. This way of writing S is useful because we may then construct it by successively constructing each term in Equation 2.14b after substituting S for Equation 2.15. By successively, we mean to group all terms of the same order in the perturbation and set the contribution from each order equal to zero individually. To be precise, H_0 is of zeroth order in this context, H_1, H_2, and $S^{(1)}$ are of first order, $S^{(2)}$ is of second order, and so on. Doing this and using that $\tilde{H}_a = 0$ and solving for commutators of form $[H_0, S^{(j)}]$ we get

$$\begin{aligned} [H_0, S^{(1)}] &= -H_2, \\ [H_0, S^{(2)}] &= -[H_1, S^{(1)}], \\ [H_0, S^{(3)}] &= -[H_1, S^{(2)}] - \frac{1}{3}[[H_2, S^{(1)}], S^{(1)}]. \end{aligned} \tag{2.16}$$

We solve Equation 2.16 for $S^{(j)}$ by looking at overlap $\langle \psi_m | \, S^{(j)} \, | \phi_l \rangle$ starting with $j = 1$ to find

$$
\begin{aligned}
S^{(1)}_{ml} &= -\frac{H'_{ml}}{E_m - E_l}, \\
S^{(2)}_{ml} &= \frac{1}{E_m - E_l}\left[\sum_{m'}\frac{H_{mm'}H_{m'l}}{E_{m'} - E_l} - \sum_{l'}\frac{H_{ml'}H_{l'l}}{E_m - E_{l'}}\right], \\
S^{(3)}_{ml} &= \frac{1}{E_m - E_l}\Bigg[-\sum_{m',m''}\frac{H_{mm'}H_{m'm''}H_{m''l}}{(E_{m'} - E_l)(E_{m''} - E_l)} \\
&\quad - \sum_{l',l''}\frac{H_{ml'}H_{l'l''}H_{l''l}}{(E_m - E_{l'})(E_m - E_{l''})} + \sum_{l',m'}\frac{H_{mm'}H_{m'l'}H_{l'l}}{(E_{m'} - E_l)(E_m - E_{l'})} \\
&\quad + \sum_{l',m'}\frac{H_{ml'}H_{l'm'}H_{m'l}}{(E_m - E_{l'})(E_{m'} - E_l)} + \frac{1}{3}\sum_{l',m'}\frac{H_{ml'}H_{l'm'}H_{m'l}}{(E_m - E_{l'})(E_{m'} - E_l)} \\
&\quad + \frac{1}{3}\sum_{l',m'}\frac{H_{ml'}H_{l'm'}H_{m'l}}{(E_m - E_{l'})(E_{m'} - E_l)} + \frac{2}{3}\sum_{l',m'}\frac{H_{ml'}H_{l'm'}H_{m'l}}{(E_m - E_{l'})(E_{m'} - E_l)}\Bigg].
\end{aligned}
\tag{2.17}
$$

Here, m's refer to states from subspace A and l's refer to states from subspace B. We have used that $H_0 \left|\psi_n\right\rangle = E_n \left|\psi_n\right\rangle$ and that $\left\langle\psi_n\right| H' \left|\psi_{n'}\right\rangle = H_{nn'}$. The final step is to insert Equation 2.17 into Equation 2.14a to get a Hamiltonian of the form

$$
H = H^{(0)} + H^{(1)} + H^{(2)} + H^{(3)} + \ldots \tag{2.18}
$$

where

$$
\begin{aligned}
H^{(0)}_{mm'} &= H^0_{mm'}, \\
H^{(1)}_{mm'} &= H'_{mm'}, \\
H^{(2)}_{mm'} &= \frac{1}{2}\sum_{l} H'_{ml}H'_{lm'}\left[\frac{1}{E_m - E_l} + \frac{1}{E_{m'} - E_l}\right], \\
H^{(3)}_{mm'} &= -\frac{1}{2}\sum_{l,m''}\left[\frac{H'_{ml}H'_{lm''}H'_{m''m'}}{(E_m - E_l)(E_{m''} - E_l)} + \frac{H'_{mm''}H'_{m''l}H'_{lm'}}{(E_m - E_l)(E_{m''} - E_l)}\right] \\
&\quad + \frac{1}{2}\sum_{l,l'} H'_{ml}H'_{ll'}H'_{l'm'} \\
&\qquad \times \left[\frac{1}{(E_m - E_l)(E_m - E_{l'})} + \frac{1}{(E_{m'} - E_l)(E_{m'} - E_{l'})}\right].
\end{aligned}
\tag{2.19}
$$

Again, indices with m refer to subspace A and l-indices refer to subspace B. The particular strength of the SWT over other approaches is that the SWT constructs a unitary transformation that conserves the Hermiticity of a Hamiltonian.

In section 3.3 we use a second-order SWT to arrive at our analytical results. To perform the SWT, we employ the PyMaBlock Python package [28, 29] in combination with SymPy for symbolic calculations. The PyMaBlock package works

by representing a Hamiltonian as a series of block matrices with the same block structure as a specified zeroth-order Hamiltonian. The algorithm then calculates, order by order, a minimal unitary matrix U that block-diagonalizes the full Hamiltonian. It does so through the minimal action principle by minimizing $||U - \mathbb{1}||$. For the case of a Hamiltonian split into two blocks, this approach is equivalent to SW.

2.3 The Luttinger-Kohn Hamiltonian

Layered semiconductor structures have garnered widespread interest in physics and technology, and the need to accurately describe such systems beyond the simple effective mass approximation has become apparent. The effective mass approximation does, for instance, not account for non-parabolicity of the band structure, spin splitting, and HH-LH coupling. A more flexible and robust approach is based on $\boldsymbol{k} \cdot \boldsymbol{p}$ theory and pseudo-degenerate perturbation theory to derive an effective Hamiltonian for the topmost valence bands and the lowest-energy conduction bands. For semiconductors with a zinc blende crystal structure, the approach may be specialized to derive Hamiltonians such as the Kane model and the LK Hamiltonian, which we are interested in. We outline in this section the derivation of the LK Hamiltonian. The approach we describe is found in [4].

The derivation starts as many calculations in quantum mechanics do: with the Schrödinger equation (SE). We write it in terms of the Bloch functions $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}u_{\nu\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\langle\boldsymbol{r}|\nu\boldsymbol{k}\rangle$ where $V(\boldsymbol{r})$ is the periodic potential of the lattice, $\boldsymbol{k}$ is the wave vector of the Bloch function, and ν denotes the energy band of the Bloch function. By evaluating the plane wave part of the Bloch functions, we may write the equation as

$$\left[\frac{p^2}{2m} + V + \frac{\hbar^2 k^2}{2m} + \frac{\hbar}{m}\boldsymbol{k}\cdot\boldsymbol{p}\right]|\nu\boldsymbol{k}\rangle = E_\nu(\boldsymbol{k})\,|\nu\boldsymbol{k}\rangle\,. \tag{2.20}$$

We then include a SOC and introduce two-component spinor versions of the Bloch functions $|n\boldsymbol{k}\rangle$ to get a Schrödinger equation that reads

$$\left[\frac{p^2}{2m_0} + V + \frac{\hbar^2 k^2}{2m} + \frac{\hbar}{m}\boldsymbol{k}\cdot\boldsymbol{\pi} + \frac{\hbar}{4m^2c^2}\boldsymbol{p}\cdot\boldsymbol{\sigma}\times(\nabla V)\right]|n\boldsymbol{k}\rangle = E_n(\boldsymbol{k})\,|n\boldsymbol{k}\rangle\,, \tag{2.21}$$

where

$$\boldsymbol{\pi} = \boldsymbol{p}\frac{\hbar}{4mc^2}\boldsymbol{\sigma}\times(\nabla V_0). \tag{2.22}$$

We now recognize that the sets of functions $\{|\nu\boldsymbol{k}\rangle\}$ and $\{|n\boldsymbol{k}\rangle\}$ are complete orthonormal bases of Equation 2.20 and Equation 2.21 respectively for a fixed wave vector $\boldsymbol{k}_0$. We may thus expand the states $|n\boldsymbol{k}\rangle$ in terms of the Bloch eigenstates at the band edge of the SE without SOC multiplied by spin states, $|\nu\boldsymbol{0}\rangle \otimes |\sigma\rangle$, as

$$|n\boldsymbol{k}\rangle = \sum_{\substack{\nu' \\ \sigma'=\uparrow,\downarrow}} c_{n\nu'\sigma}(\boldsymbol{k})\,|\nu'\sigma'\rangle\,, \quad |\nu'\sigma'\rangle := |\nu\mathbf{0}\rangle \otimes |\sigma\rangle\,. \tag{2.23}$$

We may then multiply from the left by $\langle\nu\sigma|$ and use the fact that

$$\left[\frac{p^2}{2m_0} + V_0\right]|\nu'\sigma'\rangle = E_\nu(\mathbf{0})\,|\nu'\sigma'\rangle\,, \tag{2.24}$$

to write Equation 2.21 as the eigenvalue problem

$$\sum_{\nu'\sigma'}\left[\left(E_{\nu'}(\mathbf{0})+\frac{\hbar^2k^2}{2m_0}\right)\delta_{\nu\nu'}\delta_{\sigma\sigma'}+\frac{\hbar}{m_0}\boldsymbol{k}\cdot\boldsymbol{P}^{\nu\nu'}_{\sigma\sigma'}+\Delta^{\nu\nu'}_{\sigma\sigma'}\right]c_{n\nu'\sigma'}(\boldsymbol{k}) = E_n(\boldsymbol{k})c_{n\nu\sigma}(\boldsymbol{k}), \tag{2.25}$$

where

$$\boldsymbol{P}^{\nu\nu'}_{\sigma\sigma'} \equiv \langle\nu\sigma|\boldsymbol{\pi}|\nu'\sigma'\rangle\,, \tag{2.26a}$$

$$\Delta^{\nu\nu'}_{\sigma\sigma'} := \frac{\hbar}{4m_0^2c^2}\,\langle\nu\sigma|\boldsymbol{p}\cdot\sigma\times(\nabla V_0)|\nu'\sigma'\rangle\,. \tag{2.26b}$$

If we neglect SOC, $\boldsymbol{\pi} \to \boldsymbol{p}$. When SOC is present, the degenerate energy bands are split in energy. Of particular interest to us is the uppermost valence band. Without spin, the valence band edge is threefold degenerate; with spin and SOC, the band splits into a fourfold-degenerate band with TAM 3/2 and a twofold-degenerate band with TAM 1/2. This is indeed what we see in materials such as Ge. The valence band, which has TAM 3/2, is split into two degenerate bands: a lower-energy quadruplet and a higher-energy doublet. The higher-energy doublet is referred to as the SO split-off band because SOC splits the quadruplet and doublet in energy. This gap is aptly referred to as the SO gap. In the present thesis, we will, for the sake of simplicity, only investigate the strength of the SOC in the quadruplet, thus neglecting the effect of the doublet.

One is often not interested in studying all bands of a solid-state system exactly, as this can quickly become complex, and because the dominant effects of particles in the uppermost valence band and the lowermost conduction band are not as strongly affected by bands distant in energy. One may restrict one's analysis to a finite subset of N bands by using QDPT to decouple the bands of interest from the rest of the system. In this way one may work with an N-dimensional Hamiltonian while still including the interactions from more distant bands as effective interactions. For example, one would in such a process treat the off-diagonal elements of $\boldsymbol{P}^{\nu\nu'}_{\sigma\sigma'}$ as perturbations. We will not show this procedure explicitly for the case of the Luttinger-Kohn Hamiltonian, but refer the reader to [30] where the derivation is presented in full. In such a derivation, one must necessarily assume the SOC not to be so large as to invalidate the use of perturbation theory. Out of interest, we thus highlight that the 4×4 LK Hamiltonian may be derived from symmetry considerations and by constructing invariants that obey the same symmetries as the system. This procedure may be found in [31, 4]. We note that deriving the

Hamiltonian of two different degenerate bands using the theory of invariants is not possible [4]. Thus, one would need to employ a perturbative approach for models including more bands than the four in the uppermost valence band. We present from [31] the general 4×4 LK Hamiltonian with no external fields present, which reads

$$\begin{aligned} H_{LK} =& \left(\gamma_1 + \frac{5\gamma_2}{2}\right)\frac{p^2}{2m} - \frac{\gamma_2}{m}\left(p_x^2 J_x^2 + p_y^2 J_y^2 + p_z^2 J_z^2\right) \\ & - \frac{2\gamma_3}{m}\left(\{p_x, p_y\}\{J_x, J_y\} + \{p_y, p_z\}\{J_y, J_z\} + \{p_z, p_x\}\{J_z, J_x\}\right). \end{aligned} \tag{2.27}$$

The dimensionless parameters γ_1, γ_2, and γ_3 may be either calculated perturbatively for small SOC, or inferred from experiments. Equation 2.27 is the general Luttinger-Kohn Hamiltonian for a zinc blende semiconductor with no external fields or stress applied. The Hamiltonian we present in Equation 1.1 is arrived at by imposing the so-called spherical approximation $\gamma_2 = \gamma_3 \equiv \gamma_s$. This approximation does not hold for any zinc blende semiconductor, only ones where $\gamma_2 \sim \gamma_3$. In Ge $\gamma_2 \approx 4.24$ and $\gamma_3 \approx 5.69$, which are close enough for the approximation $\gamma_s = (\gamma_2 + \gamma_3)/2$ to give acceptable results, though we discuss this choice in section 5.1. In this case we may rewrite Equation 2.27 as

$$H_{LK}^s = \left(\gamma_1 + \frac{5\gamma_s}{2}\right)\frac{p^2}{2m} - \frac{\gamma_s}{m}\left(\boldsymbol{p}\cdot\boldsymbol{J}\right)^2, \tag{2.28}$$

where we used the identity [31]

$$\{p_\alpha, p_\beta\}\{J_\alpha, J_\beta\} = \left(\boldsymbol{p}\cdot\boldsymbol{J}\right)^2 - \frac{e}{2c}(\boldsymbol{J}\cdot\boldsymbol{B}). \tag{2.29}$$

We have used ESC in Equation 2.29, where $\boldsymbol{B}$ is a magnetic field whose term is not included in Equation 2.28.

2.4 Bir-Pikus Strain Model

Strain in a semiconductor can drastically alter the energy spectrum of a material. Most notably for the case of uniaxial strain, meaning along one axis, in Ge the HHs and LHs are split in energy at the previously degenerate Γ-point. The framework for accounting for strain in our model is based upon the work of Bir and Pikus (1974) [32] and presented explicitly for strained Ge by Terrazos et al. (2021) [33]. Though we will not present a full derivation of the strain term, we will outline how it arises, how it is incorporated in the LK Hamiltonian, and some of the relevant parameters defining the strength of the strain in the system.

Strain may be introduced near interfaces between two coherently grown microscale planar, or nanoscale spherical or cylindrical core/shell heterostructures because of the differences in lattice constants and thermal properties [34]. Such strain may be compressive or tensile based on the materials used. What this in effect means for the material is that the crystal structure is warped in such a way that the two

materials try to obtain an equilibrium also near the interface between the differing materials.

The big problem to overcome when incorporating strain into the Hamiltonian is to realize that the spatial symmetry group of the strained material is generally not a subgroup of the unstrained symmetry group. For this to be the case, one must introduce a "deformed" coordinate system via a strain tensor ε_{ij} such that the strained coordinates are

$$r_i' = \sum_j (\delta_{ij} - \varepsilon_{ij}) r_j, \tag{2.30a}$$

$$p_i' = \sum_j (\delta_{ij} + \varepsilon_{ij}) p_j, \tag{2.30b}$$

$$k_i' = \sum_j (\delta_{ij} + \varepsilon_{ij}) k_j. \tag{2.30c}$$

In this coordinate system, the strained lattice points coincide with the unstrained lattice. One may then derive an energy splitting by decomposing the IRs of the symmetry group of the unstrained lattice in terms of the IRs of the strained lattice's symmetry group [4, 32].

In [33] the authors present the 6×6 LKBP Hamiltonian, i.e. the LK Hamiltonian with strain incorporated to first order. For the system we are interested in, described in section 3.1, we will impose the special case of uniaxial strain along the z-direction. One thus assumes that the material is grown in a high symmetry direction and with no sources of shear strain such that $\varepsilon_{xy} = \varepsilon_{yz} = \varepsilon_{zx} = 0$. One also assumes that the strain in the xy-plane is uniform and thus $\varepsilon_{xx} = \varepsilon_{yy} \equiv \varepsilon_{\parallel}$. Neglecting the SO split-off band to obtain the 4×4 LKBP Hamiltonian we are interested in, one finds that, under these conditions, the strain appears only on the diagonal in the TAM 3/2 basis. The result is that the term

$$H_\varepsilon = |b| \varepsilon_0 J_z^2 \tag{2.31}$$

is added to the LK Hamiltonian to get the LKBP Hamiltonian. In Equation 2.31, $b\varepsilon_0$ is the Bir-Pikus strain energy [24, 32], and for strained Ge we have that $b = -2.16$ eV [35],

$$\varepsilon_0 \equiv \varepsilon_{\parallel} - \varepsilon_{zz} \approx 1.74 \varepsilon_{\parallel}, \tag{2.32}$$

since

$$\varepsilon_{zz} = -\frac{2C_{12}\varepsilon_{\parallel}}{C_{11}} \approx -0.74 \varepsilon_{\parallel}, \tag{2.33}$$

where C_{ij} are the elastic constants for Ge (whose values may be found in [35], though we will not use their values in the present thesis).

CHAPTER

THREE

METHODS

In this chapter, we present the approaches we have employed to obtain our results. In section 3.1, we describe the system we are investigating as well as its Hamiltonian. Further, in section 3.2, we present the numerical approach we used to diagonalize our Hamiltonian, and, finally, in section 3.3, we present the steps we took to obtain our analytical results.

3.1 Setting up the Problem

We wish to investigate the strength of the SOC of the lowest-energy Kramers pair states of a TAM 3/2 hole in a quasi-1D wire in uniaxially strained Ge. Such a system may be constructed using a thin layer of Ge sandwiched between two layers of $Si_{1-x}Ge_x$, where x here refers to the concentration of each atom in the crystal. If the layers are stacked in the z-direction and the Ge layer, whose width we denote w_z, is thin enough, the holes will be confined to the ground state in the z-direction. The strain will also then lie along the z-direction. We also restrict the motion of the hole in the x-direction by imposing a hard-wall potential of width w_x in that direction. We denote the confining potential composed of the hard-wall potentials in the x- and z-direction by

$$V_c = V_c(x,z) = \begin{cases} 0, & (0 < x < w_x) \wedge (0 < z < w_z) \\ \infty, & \text{else} \end{cases} . \tag{3.1}$$

For the present thesis, the hole is left unrestricted in the y-direction, thus creating an effective 1D channel in the y-direction. We then add an electric field E along the z-direction to localize the hole near the topmost interface and to induce an E-dependent SOC. The localization of the hole near the top boundary is characterized by a characteristic electric length $l_E \equiv (\hbar^2\gamma_1/2meE)^{1/3} \approx 8\text{nm} \times E^{-1/3}$, where E is given in V/µm [36]. A sketch of the system is provided in Figure 3.1.1.

Such a system can be modeled by the LKBP Hamiltonian with additional electric terms added, thus taking the form

$$H_{\text{LKBP}} = \left(\gamma_1 + \frac{5\gamma_s}{2}\right)\frac{\boldsymbol{p}^2}{2m} - \frac{\gamma_s}{m}(\boldsymbol{p}\cdot\boldsymbol{J})^2 + |b|\varepsilon_0 J_z^2 + V_c - eEz, \tag{3.2}$$

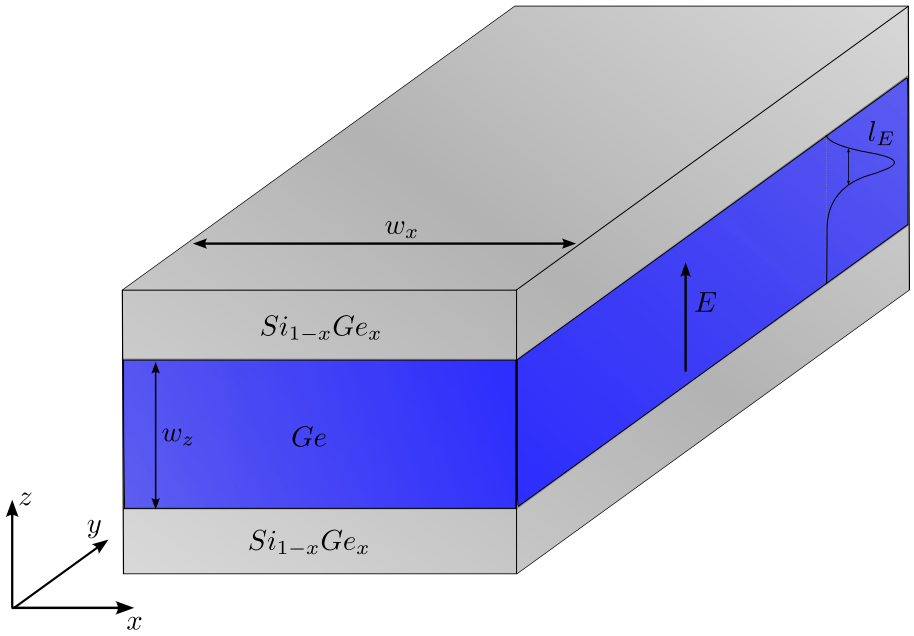


Figure 3.1.1: A sketch of the system we investigate in the present thesis. We imagine a planar heterostructure in which holes are confined in a layer of Ge sandwiched between two layers of $Si_{1-x}Ge_x$. The thickness of the Ge layer is denoted by w_z, and an electric field E is applied along the positive z direction. This electric field localizes the hole's wave function near the upper boundary of the heterostructure with a characteristic electric length l_E. The hole is confined in the x-direction by a hard-wall potential of width w_x and is free in the y-direction.

where m is the electron mass, $\boldsymbol{p} = -i\hbar\nabla$ is the canonical momentum operator, and $\boldsymbol{J} = (J_x, J_y, J_z)$ is the vector of spin-3/2 matrices. The confining electrostatic field $V_c = V_c(x, z)$ comprises a rectangular hard wall potential in the (x, z) plane. Furthermore, γ_1 and γ_s are the isotropic Luttinger-Kohn (LK) parameters. For Ge these have values $\gamma_1 \approx 13.35$ and we have used $\gamma_s \equiv (\gamma_2 + \gamma_3)/2 \approx 4.96$. These parameters quantify the mixing between the heavy hole (HH) and light hole (LH) states of our system, which are further split by the Bir-Pikus strain energy $b\varepsilon_0$. By HH, we mean the TAM 3/2 hole states with $m_j = 3/2$, and by light holes, we mean the TAM 3/2 hole states with $m_j = 1/2$. We refer to these states as "heavy" and "light" because the HHs have a greater effective mass in the plane perpendicular to the strain axis, and the LHs have a smaller effective mass.

For future convenience we make the Hamiltonian in 3.2 dimensionless by introducing the dimensionless operators:

$$\tilde{r}_i \equiv \frac{r_i}{L}, \qquad \tilde{p}_i \equiv \frac{L}{\hbar} p_i, \qquad \tilde{H}_{\text{LKBP}} \equiv \frac{H_{\text{LKBP}}}{\hbar^2\gamma_1/2mL^2}, \tag{3.3}$$

where L is an arbitrary length scale of our choosing and m/γ_1 is the average effective mass of the HHs and the LHs. With this, we get the dimensionless Hamiltonian operator

$$\tilde{H}_{\text{LKBP}} = \left(1 + \frac{5\gamma_s}{2\gamma_1}\right)\tilde{\boldsymbol{p}}^2 - \frac{2\gamma_s}{\gamma_1}(\tilde{\boldsymbol{p}} \cdot \boldsymbol{J})^2 + \tilde{\varepsilon}_0 J_z^2 + V_c - \tilde{E}\tilde{z}, \tag{3.4}$$

where $\tilde{\varepsilon}_0 \equiv \varepsilon_0/\varepsilon_E = \varepsilon_0/\frac{\hbar^2\gamma_1}{2m|b|L^2}$ and $\tilde{E} \equiv E/\frac{\hbar^2\gamma_1}{2meL^3}$. In future expressions, we shall omit the tilde when referring to dimensionless parameters, and we will use $L = l_E$ to present our results unless otherwise specified.

3.2 Numerics

To investigate the system described in section 3.1, we discretize our Hamiltonian in the xz-plane and employ a numerical approach. To do this, we create a rectangular lattice of points that are evenly spaced in each direction by distances a_x and a_z. The xz-plane of the system has a cross-sectional area of $w_x \times w_z$ where $w_x = a_x(N_x+1)$ and $w_z = a_z(N_z+1)$. Here N_α denotes the number of lattice sites in the r_α-direction. Denote the positions of the lattice sites by $(x_i, z_j) = (ia_x, ja_z)$ where $(i,j) \in I_x \times I_z$ and $I_\alpha \equiv \{1, 2, \dots N_\alpha\}$. Notice here that $r_\alpha \in \{a_\alpha, 2a\alpha, \dots, w_\alpha - a\alpha\}$ for $\alpha = x, z$. This is done to correctly implement the hard-wall potential in the discrete case, ensuring that the hard walls occur at $x = 0$ or $x = w_x$, or at $z = 0$ or $z = w_z$. With this in mind, we get the discretized position operators

$$(\underline{r_\alpha})_{i,j} = \langle (r_\alpha)_j | r_\alpha | (r_\alpha)_i \rangle = (r_\alpha)_j \delta_{i,j}. \tag{3.5}$$

Here, $|(r_\alpha)_i\rangle$ refers to the quantum state of the system where the hole is localized at the ith lattice site at position $(r_\alpha)_i = ia_\alpha$ in the α-direction. Motion in this discrete model constitutes hopping between neighboring lattice sites. To describe the hopping, we discretize the partial differentiation operators ∂_α using finite differences. This yields discretized operators

$$\partial_\alpha \to \frac{|(r_\alpha)_{i+1}\rangle\langle(r_\alpha)_i| - |(r_\alpha)_i\rangle\langle(r_\alpha)_i|}{a_\alpha}. \tag{3.6}$$

Simultaneously, the operator may also be written as

$$\partial_\alpha \to \frac{|(r_\alpha)_{i-1}\rangle\langle(r_\alpha)_i| - |(r_\alpha)_i\rangle\langle(r_\alpha)_i|}{-a_\alpha}. \tag{3.7}$$

Taking the average of Equation 3.6 and Equation 3.7 we get that

$$\partial_\alpha \to \frac{1}{2a_\alpha}\big(|(r_\alpha)_{i+1}\rangle\langle(r_\alpha)_i| - |(r_\alpha)_{i-1}\rangle\langle(r_\alpha)_i|\big), \tag{3.8}$$

which finally results in the discretized canonical momentum operator on the lattice being

$$(\underline{p_\alpha})_{ij} \equiv \langle (r_\alpha)_j | p_\alpha | (r_\alpha)_i \rangle = -\frac{i}{2a_\alpha}\big(\delta_{i,j+1} - \delta_{i,j-1}\big). \tag{3.9}$$

Similarly, we may derive the second-order derivative and thus the discretized squared canonical momentum operator to be

$$(\underline{p_\alpha^2})_{i,j} \equiv \langle (r_\alpha)_j | p_\alpha^2 | (r_\alpha)_i \rangle = -\frac{1}{a_\alpha^2}(\delta_{i,j+1} + \delta_{i,j-1} - 2\delta_{i,j}). \tag{3.10}$$

The full Hilbert space is a direct product between the spatial subspace and the TAM subspace. We may thus get discretized operators acting on our full Hilbert space by taking Kronecker products of spatial operators and TAM operators. To make working with quantum states, tensor products, and partial tracing simpler, we have used the QuTiP Python package to write our code. [37, 38, 39]

We use the method outlined in [25] to calculate the spin-orbit velocity v. We rearrange our Hamiltonian to the form $H = H_0 + H_1 p_y + H_2 p_y^2$. We have that $H_0 = H_{\text{LKBP}}|_{p_y=0}$. We then diagonalize H_0 by finding a matrix M_E such that $M_E^\dagger H_0 M_E$ is diagonal. We then pick out the two lowest-energy states $|0\rangle^{(0)}$ and $|1\rangle^{(0)}$ and project H_1 onto the subspace spanned by these two states, yielding the first-order perturbative SOC term

$$(\boldsymbol{v}\cdot\boldsymbol{\sigma})_{ij} = (M_E^\dagger H_1 M_E)_{ij} = \langle j|^{(0)} H_1 |i\rangle^{(0)}; \quad i,j = 0,1. \tag{3.11}$$

Here, $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the vector with the Pauli spin matrices as components and $\boldsymbol{v} \equiv (v_x, v_y, v_z)$ is the spin-orbit velocity vector. Since our Hamiltonian is time-reversal symmetric, the two lowest energy states will be a degenerate Kramers pair per Kramers' degeneracy theorem [40].

To verify that the numerics we develop are adequate, we attempt to reproduce some of the results presented in [24], namely plots (a) and (d) in FIG. 2. In [24], they investigate a very similar system to ours, only that the authors introduced a harmonic potential $\hbar^2\gamma_1 x^2/2ml_x^4$ to the Hamiltonian of the system. Plot (a) of FIG. 2 in the paper shows the SOV for some fixed widths w_z as functions of the strength of the harmonic potential in the x-direction characterized by the length l_x. The second plot shows how the SOV varies with the confinement length l_x and the strain parameter ε_0. Our reproduction is presented in Figure 3.2.1, where Figure 3.2.1a corresponds to plot FIG. 2(a) in [24] and Figure 3.2.1b corresponds to FIG. 2(d). Since the magnitude and shape of the plots we produce match those in [24] to a satisfying degree, we may be confident that our numerics are up to par.

3.3 Perturbative Approach

We attempt to derive a perturbative approximate model for the system described in 3.1 when the electric field E is small. We do this with a similar approach to the one used in Appendix B and C of [25]; by projecting the LKBP Hamiltonian

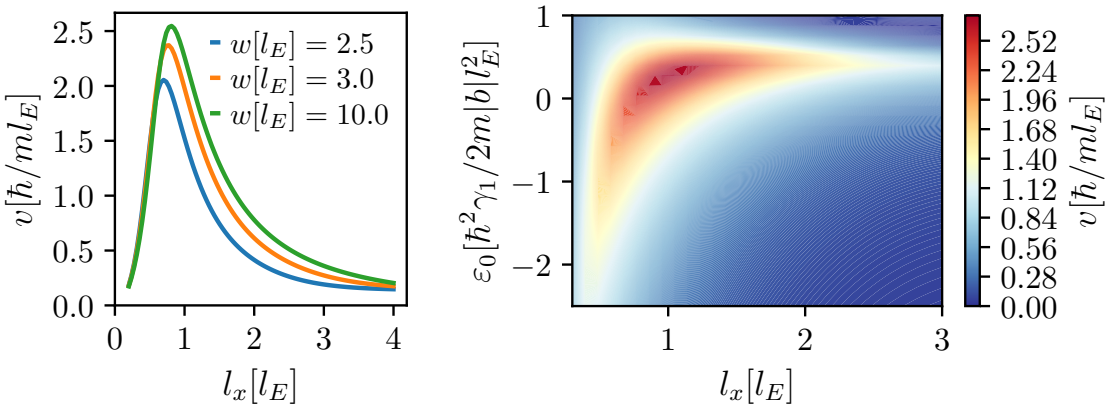


Figure 3.2.1: A reproduction of Figure 2a (here, left plot) and Figure 2d (here, right plot) presented in [24] to confirm the correctness of our numerical implementation. The left plot shows the amplitude of the spin-orbit velocity for three values of the confinement in the z-direction and as a function of the strength of the harmonic confinement in the x-direction, characterized by the length l_x. The right plot shows the spin-orbit velocity for $w_z = 5$ as a function of the strain parameter ε_0 and l_x. All v's are given in units of $\hbar/ml_E$, ε_0 in units of $\hbar^2\gamma_1/2m|b|l_E^2$, and all lengths in units of l_E.

onto a subset of the eigenbasis of a 2D particle-in-a-box (PIAB) Hamiltonian and deriving an effective wire Hamiltonian using a SWT. The last step is done similarly to the numerical approach, where we rewrite our Hamiltonian in the PIAB basis to be of the form

$$\underline{H} = \underline{H_0} + \underline{H_1(p_y)} + p_y^2\underline{H_2}. \tag{3.12}$$

The difference from our numerical approach is that we cannot analytically diagonalize $H_{\text{LKBP}}|_{p_y=0}$, and must thus treat some of the terms as perturbations by including them in $\underline{H_1(p_y)}$. We will also perform a second-order SWT, and not only a first-order one.

Since we want to integrate out the xz-dimensions of the LKBP Hamiltonian, we set $p_y = 0$ and work only in the xz-plane for now. We denote the eigenbasis of the 2D particle-in-a-box Hamiltonian by $\{|n_x, n_z\rangle\}_{n_x=1,n_z=1}^{\infty}$. The eigenfunctions of the 2D particle-in-a-box Hamiltonian are

$$\langle x, z|n_x, n_z\rangle = \Psi_{n_x,n_z}(x,z) = \frac{2}{\sqrt{w_x w_z}} \sin\left(\frac{\pi n_x}{w_x}x\right) \sin\left(\frac{\pi n_z}{w_z}z\right), \tag{3.13}$$

when $0 < x < w_x$ and $0 < z < w_z$. We project the coordinate and momentum operators onto this basis to get

$$\begin{aligned}\underline{r_\alpha} &= \sum_{n_x,\nu_x,n_z,\nu_z} \langle \nu_x,\nu_z|r_\alpha|n_x,n_z\rangle|\nu_x,\nu_z\rangle\langle n_x,n_z|,\\ \underline{p_{\alpha\alpha'}} &= \sum_{n_x,\nu_x,n_z,\nu_z} \langle \nu_x,\nu_z|p_\alpha p_{\alpha'}|n_x,n_z\rangle|\nu_x,\nu_z\rangle\langle n_x,n_z|,\end{aligned} \tag{3.14}$$

where, for an operator $\hat{\mathcal{O}}$,

$$\langle \nu_x,\nu_z|\hat{\mathcal{O}}|n_x,n_z\rangle = \iint dxdz\, \Psi_{\nu_x,\nu_z}(x,z)\hat{\mathcal{O}}\Psi_{n_x,n_z}(x,z). \tag{3.15}$$

We may calculate the projected operators analytically to be

$$\big(\underline{p_{xx}}\big)_{n_x,n_z;\nu_x,\nu_z} = \left(\frac{\pi n_x}{w_x}\right)^2 \delta_{n_x,\nu_x}\delta_{n_z,\nu_z}, \tag{3.16a}$$

$$\big(\underline{p_{zz}}\big)_{n_x,n_z;\nu_x,\nu_z} = \left(\frac{\pi n_z}{w_z}\right)^2 \delta_{n_x,\nu_x}\delta_{n_z,\nu_z}, \tag{3.16b}$$

$$\big(\underline{p_{xz}}\big)_{n_x,n_z;\nu_x,\nu_z} = -\frac{4\big(1-(-1)^{n_x+\nu_x}\big)\big(1-(-1)^{n_z+\nu_z}\big)n_x n_z \nu_x \nu_z}{(\nu_x^2-n_x^2)(\nu_z^2-n_z^2)w_x w_z}, \tag{3.16c}$$

$$\big(\underline{p_{xy}}\big)_{n_x,n_z;\nu_x,\nu_z} = p_y \frac{2i\big(1-(-1)^{n_x+\nu_x}\big)n_x\nu_x}{w_x(\nu_x^2-n_x^2)}\delta_{n_z,\nu_z}, \tag{3.16d}$$

$$\big(\underline{p_{yz}}\big)_{n_x,n_z;\nu_x,\nu_z} = p_y \frac{2i\big(1-(-1)^{n_z+\nu_z}\big)n_z\nu_z}{w_z(\nu_z^2-n_z^2)}\delta_{n_x,\nu_x}, \tag{3.16e}$$

$$(\underline{x})_{n_x,n_z;\nu_x,\nu_z} = \frac{4\big(1-(-1)^{n_x+\nu_x}\big)n_x\nu_x w_x}{\pi^2\big(\nu_x^2-n_x^2\big)^2}\delta_{n_z,\nu_z}, \tag{3.16f}$$

$$(\underline{z})_{n_x,n_z;\nu_x,\nu_z} = \frac{4\big(1-(-1)^{n_z+\nu_z}\big)n_z\nu_z w_z}{\pi^2\big(\nu_z^2-n_z^2\big)^2}\delta_{n_x,\nu_x}. \tag{3.16g}$$

We have in $\underline{z}$ discarded any irrelevant shift of all energies. Inserting these projected operators into Equation 3.2, we get the projected zeroth-order in p_y Hamiltonian

$$\begin{aligned}\underline{H_{\text{LKBP}}}|_{p_y=0} =& \left[\left(\gamma_1+\frac{5\gamma_s}{2}\right)(\underline{p_{xx}}+\underline{p_{zz}}) + V_c\mathbb{1} - E\underline{z}\right]\otimes\mathbb{1}_{4\times4} + \varepsilon_0\mathbb{1}\otimes J_z^2\\ &- 2\gamma_s\big(\underline{p_{xx}}\otimes J_x^2 + \underline{p_{zz}}\otimes J_z^2 + \underline{p_{xz}}\otimes\{J_x,J_z\}\big),\end{aligned} \tag{3.17}$$

where $\otimes$ is a Kronecker product. In order to perform a SWT we must have a diagonal matrix as an unperturbed Hamiltonian. We would ideally diagonalize Equation 3.17 as we did in section 3.2 and work in the diagonal basis, but we have not found a useful way of doing this. We must thus construct a zeroth-order Hamiltonian $\underline{H_0}$ that we are able to easily diagonalize, by carefully choosing what

terms from Equation 3.17 to include. The terms we do not include in our zeroth-order Hamiltonian we treat as perturbations by adding them to the Hamiltonian $\underline{H_1}$. In this process, we must then make sure that the terms we leave as perturbations are not big relative to the relevant energy splittings.

We find that the Hamiltonian

$$\underline{H_0} = \left(1 + \frac{5\gamma_s}{2\gamma_1}\right)(\underline{p_{xx}} + \underline{p_{zz}}) \otimes \mathbb{1}_{4\times4} - \frac{2\gamma_s}{\gamma_1}\left(\underline{p_{xx}} \otimes J_x^2 + \underline{p_{zz}} \otimes J_z^2\right) + \tilde{\varepsilon}_0 \mathbb{1} \otimes J_z^2, \quad (3.18)$$

is straightforward to diagonalize in 2×2 blocks. This choice also ensures that couplings due to the term $-(2\gamma_s/\gamma_1)\underline{p_{xx}} \otimes J_x^2$ do not cause problems for our perturbation theory, as they are, for some values of w_x and w_z, very large compared to the energy splitting of the relevant states. The remaining terms from Equation 3.17 we include in the perturbation Hamiltonian, which, after no longer setting p_y to zero, reads

$$\underline{H_1} = -\frac{2\gamma_s}{\gamma_1}\Big(\underline{p_{xz}} \otimes \{J_x, J_z\} + \underline{p_{xy}} \otimes \{J_x, J_y\} + \underline{p_{yz}} \otimes \{J_y, J_z\}\Big) - \tilde{E}\underline{z} \otimes \mathbb{1}_{4\times4}. \quad (3.19)$$

We assume here that the electrostatic field E is small and that we may indeed treat it perturbatively. Note also that we include the term $H_{xz} \equiv -(2\gamma_s/\gamma_1)\underline{p_{xz}} \otimes \{J_x, J_z\}$ in Equation 3.19. This term couples HHs and LHs with different parity quantum numbers in the xz-plane. These couplings are also, for some values of w_x and w_z, somewhat large compared to the relevant energy splittings. For the holes with $n_x = n_z = 1$, however, the only potentially problematic couplings are to the states with $n_x = n_z = 2$. We discuss our choice of $\underline{H_0}$ and $\underline{H_1}$ and the validity of the perturbation theory we perform further in section 5.1.

For the sake of simplifying notation, we rewrite Equation 3.18 as

$$\underline{H_0} = \begin{pmatrix} B_{1,1} & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{0} & B_{1,2} & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{0} & B_{2,1} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (3.20)$$

Note that $\underline{H_0}$ is a block-diagonal matrix where we have defined

$$B_{n_x,n_z} \equiv \begin{pmatrix} n_\alpha^2\varepsilon_\alpha^- + \frac{9\varepsilon_0}{4} & 0 & 0 & -\frac{n_x^2 t_{xx}}{2} \\ 0 & n_\alpha^2\varepsilon_\alpha^- + \frac{9\varepsilon_0}{4} & -\frac{n_x^2 t_{xx}}{2} & 0 \\ 0 & -\frac{n_x^2 t_{xx}}{2} & n_\alpha^2\varepsilon_\alpha^+ + \frac{\varepsilon_0}{4} & 0 \\ -\frac{n_x^2 t_{xx}}{2} & 0 & 0 & n_\alpha^2\varepsilon_\alpha^+ + \frac{\varepsilon_0}{4} \end{pmatrix}. \quad (3.21)$$

In Equation 3.21, repeated indices $\alpha = x, y$ are summed over. We have further defined the parameters

$$\varepsilon_\alpha^\pm \equiv \frac{\pi^2}{m_\alpha^\pm w_\alpha^2}, \quad \frac{1}{m_x^\pm} \equiv 1 \mp \frac{\gamma_s}{\gamma_1}, \quad \frac{1}{m_z^\pm} \equiv 1 \pm \frac{2\gamma_s}{\gamma_1}, \quad t_{xx} \equiv \frac{2\sqrt{3}\pi^2\gamma_s}{w_x^2\gamma_1}. \tag{3.22}$$

The next step is to choose a finite subset of the PIAB eigenbasis to work in. We choose to work with the three basis states $|1,1\rangle$, $|1,2\rangle$, and $|2,1\rangle$, and consequently we will work in the subspace $\mathcal{H}_{12}$ which is spanned by the basis

$$\begin{aligned}\Big\{ &|1,1,3/2\rangle, |1,1,-3/2\rangle, |1,1,1/2\rangle, |1,1,-1/2\rangle, \\ &|1,2,3/2\rangle, |1,2,-3/2\rangle, |1,2,1/2\rangle, |1,2,-1/2\rangle, \\ &|2,1,3/2\rangle, |2,1,-3/2\rangle, |2,1,1/2\rangle, |2,1,-1/2\rangle \Big\},\end{aligned} \tag{3.23}$$

where we define $|n_x, n_z, m_j\rangle \equiv |n_x, n_z\rangle \otimes |m_j\rangle$ to be the vectors which are constructed by taking a direct product between PIAB and TAM 3/2 basis vectors. Here we are also using that $|m_j\rangle = |3/2, m_j\rangle$ are the TAM 3/2 eigenstates where $m_j = \pm 3/2, \pm 1/2$ is the projection of the TAM along the z-direction. We denote the submatrix of Equation 3.20 and Equation 3.19 in this basis by $\underline{H_0^{12}}$ and $\underline{H_1^{12}}$ respectively.

Further, we want to diagonalize $\underline{H_0^{12}}$. As mentioned before, $\underline{H_0^{12}}$ can be straightforwardly diagonalized in 2×2 blocks. This is evident from the structure of the blocks on the diagonal given in Equation 3.21, where each diagonal element is coupled only to one other diagonal element. We construct a matrix $\underline{R}$ such that $\underline{D_0} \equiv \underline{R}^T \underline{H_0^{12}} \underline{R}$ is a diagonal matrix. The matrix $\underline{R}$ will have the same block-diagonal form as $\underline{H_0^{12}}$, and each 2×2 block is given by a rotation matrix. From Appendix 6 we get that

$$\underline{R} = \begin{pmatrix} R_{1,1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & R_{1,2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & R_{2,1} \end{pmatrix}, \tag{3.24}$$

where

$$R_{n_x,n_z} = \begin{pmatrix} \beta^+_{n_x,n_z} & 0 & 0 & -\beta^-_{n_x,n_z} \\ 0 & \beta^+_{n_x,n_z} & -\beta^-_{n_x,n_z} & 0 \\ 0 & \beta^-_{n_x,n_z} & \beta^+_{n_x,n_z} & 0 \\ \beta^-_{n_x,n_z} & 0 & 0 & \beta^+_{n_x,n_z} \end{pmatrix}, \tag{3.25}$$

and

$$\begin{aligned} &\beta^\pm_{n_x,n_z} \equiv \frac{1}{\sqrt{2}}\sqrt{1 \pm \frac{|\Delta_{n_x,n_z}|}{\omega_{n_x,n_z}}}, \quad \omega_{n_x,n_z} \equiv \sqrt{\Delta^2_{n_x,n_z} + n_x^4 t_{xx}^2}, \\ &\Delta_{n_x,n_z} \equiv n_x^2\Delta_x + n_z^2\Delta_z - 2\tilde{\varepsilon}_0, \quad \Delta_x \equiv \varepsilon_x^+ - \varepsilon_x^- = -\frac{2\pi^2\gamma_s}{w_x^2\gamma_1}, \\ &\Delta_z \equiv \varepsilon_z^+ - \varepsilon_z^- = \frac{4\pi^2\gamma_s}{w_z^2\gamma_1}, \end{aligned} \tag{3.26}$$

and where the last parameters are defined in Equation 3.22. Using Equation 3.20, Equation 3.21, Equation 3.24, and Equation 3.25 we find that

$$(\underline{D_0})_{n_x,n_z,s_1;\nu_x,\nu_z,s_2} \equiv \delta_{n_x,\nu_x}\delta_{n_z,\nu_z}\delta_{s_1,s_2}\begin{cases}\varepsilon_{n_x,n_z} - \frac{1}{2}\omega_{n_x,n_z}, & s_1 = 1,2\\ \varepsilon_{n_x,n_z} + \frac{1}{2}\omega_{n_x,n_z}, & s_1 = 3,4\end{cases}, \tag{3.27}$$

where we have defined

$$\varepsilon_{n_x,n_z} \equiv \frac{1}{4}\Big[2\big(n_x^2\varepsilon_x + n_z^2\varepsilon_z\big) + 5\tilde{\varepsilon}_0\Big], \tag{3.28}$$

and where $|n_x, n_z, s\rangle \equiv \underline{R}\,|n_x, n_z, m_j\rangle$ are the eigenstates of $\underline{D_0}$. Note that $|n_x, n_z, 1\rangle$ and $|n_x, n_z, 2\rangle$, and $|n_x, n_z, 3\rangle$ and $|n_x, n_z, 4\rangle$, are both Kramers pairs and thus pairwise degenerate; which is apparent from Equation 3.27.

The next step is to change the basis of $\underline{H_1^{12}}$, the submatrix one gets by projecting Equation 3.19 onto the basis given in Equation 3.23, to the eigenbasis of Equation 3.20. This is done by calculating the projected perturbation Hamiltonian $\underline{D_1} \equiv \underline{R}^T\underline{H_1^{12}}\underline{R}$. Finally, to derive the SOV from $\underline{D_1}$, we perform a second order SWT, using $\mathrm{span}\big(\{|1,1,1\rangle, |1,1,2\rangle\}\big)$ as the low-energy subspace. Let H_W^1 be the resulting submatrix one gets by projecting the transformed $\underline{D_1}$ onto the lowest-energy Kramers pair. Then we have that

$$H_W^1 = p_y\boldsymbol{v}\cdot\boldsymbol{\sigma}, \quad v_\alpha = \frac{1}{p_y}\mathrm{tr}\big(H_W^1\sigma_\alpha\big). \tag{3.29}$$

In chapter 4 we give an explicit expression for v, as well as compare our analytical results to numerical calculations.

CHAPTER

FOUR

RESULTS

In the following chapter, we present our results. In section 4.1, we present the explicit expression. Further, we compare the analytical expression with numerical results in section 4.2 to assess the accuracy of our calculation.

4.1 Explicit Spin-Orbit Velocity

Using the method we describe in section 3.3, we derive a perturbative expression for the SOV of the system that reads

$$v = 4t_E t_z \omega_{1,2} \left| \frac{(\beta^-_{1,2}\beta^+_{1,1} - \beta^-_{1,1}\beta^+_{1,2})(\beta^-_{1,1}\beta^-_{1,2} + \beta^+_{1,1}\beta^+_{1,2})}{(\omega_{1,1} + 6\pi^2/w_z^2)^2 - \omega^2_{1,2}} \right|. \tag{4.1}$$

Summarizing the definitions of the variables and defining the last one, we have that

$$\begin{aligned}
&\beta^{\pm}_{n_x,n_z} \equiv \frac{1}{\sqrt{2}}\sqrt{1 \pm \frac{|\Delta_{n_x,n_z}|}{\omega_{n_x,n_z}}}, \quad \omega_{n_x,n_z} \equiv \sqrt{\Delta^2_{n_x,n_z} + n_x^4 t_{xx}^2}, \\
&\Delta_{n_x,n_z} \equiv n_x^2\Delta_x + n_z^2\Delta_z - 2\varepsilon_0, \quad \Delta_x \equiv \varepsilon_x^+ - \varepsilon_x^- = -\frac{2\pi^2\gamma_s}{w_x^2\gamma_1}, \\
&\Delta_z \equiv \varepsilon_z^+ - \varepsilon_z^- = \frac{4\pi^2\gamma_s}{w_z^2\gamma_1}, \quad \varepsilon_{n_x,n_z} \equiv \frac{1}{4}\Big[2\big(n_x^2\varepsilon_x + n_z^2\varepsilon_z\big) + 5\varepsilon_0\Big], \\
&\varepsilon_\alpha \equiv \varepsilon_\alpha^+ + \varepsilon_\alpha^- = \frac{2\pi^2}{w_\alpha^2}, \quad \varepsilon_\alpha^\pm \equiv \frac{\pi^2}{m_\alpha^\pm w_\alpha^2}, \quad \frac{1}{m_x^\pm} \equiv 1 \mp \frac{\gamma_s}{\gamma_1}, \quad \frac{1}{m_z^\pm} \equiv 1 \pm \frac{2\gamma_s}{\gamma_1} \\
&t_{xx} \equiv \frac{2\sqrt{3}\pi^2\gamma_s}{w_x^2\gamma_1}, \quad t_z \equiv \frac{32\gamma_s}{\sqrt{3}w_z\gamma_1}, \quad t_E \equiv \frac{32 w_z \tilde{E}}{9\pi^2}.
\end{aligned} \tag{4.2}$$

Equation 4.1 is an explicit expression for the SOV. From it, it is apparent that the only excited state that contributes to the SOV in our truncated subspace to second order is the one for which $n_x = 1$ and $n_z = 2$, and that the effective SOC is mediated by a virtual state which couples to one of the ground states by a term $\propto t_z p_y$ and to the other by a term $\propto t_E$. Also, we find that the effective SOC between the two lowest-energy states of the system aligns with the x-direction.

4.2 Comparing Analytical and Numerical Results

With this, we compare our analytic expression with numerical results to evaluate the accuracy of our results. In Figure 4.2.1 we have presented numerical and analytical SOV distributions as functions of w_x for $\varepsilon_0 = 0$ and four different values of w_z. What we can see from these plots is that for values of w_z that are smaller or similar to l_E and for $w_x \sim w_z$, our analytical expression captures relatively well the magnitude and shape of the SOV. In the regime where $w_x < w_z$, however, the analytical expression fails to capture the magnitude of the SOV. For $w_x > w_z$, the analytical expression matches the shape of the numerical plot quite well, and the magnitude matches better than for $w_x < w_z$, though the relative difference in magnitude does differ significantly and seems to saturate at around 40% as $w_x \to \infty$.

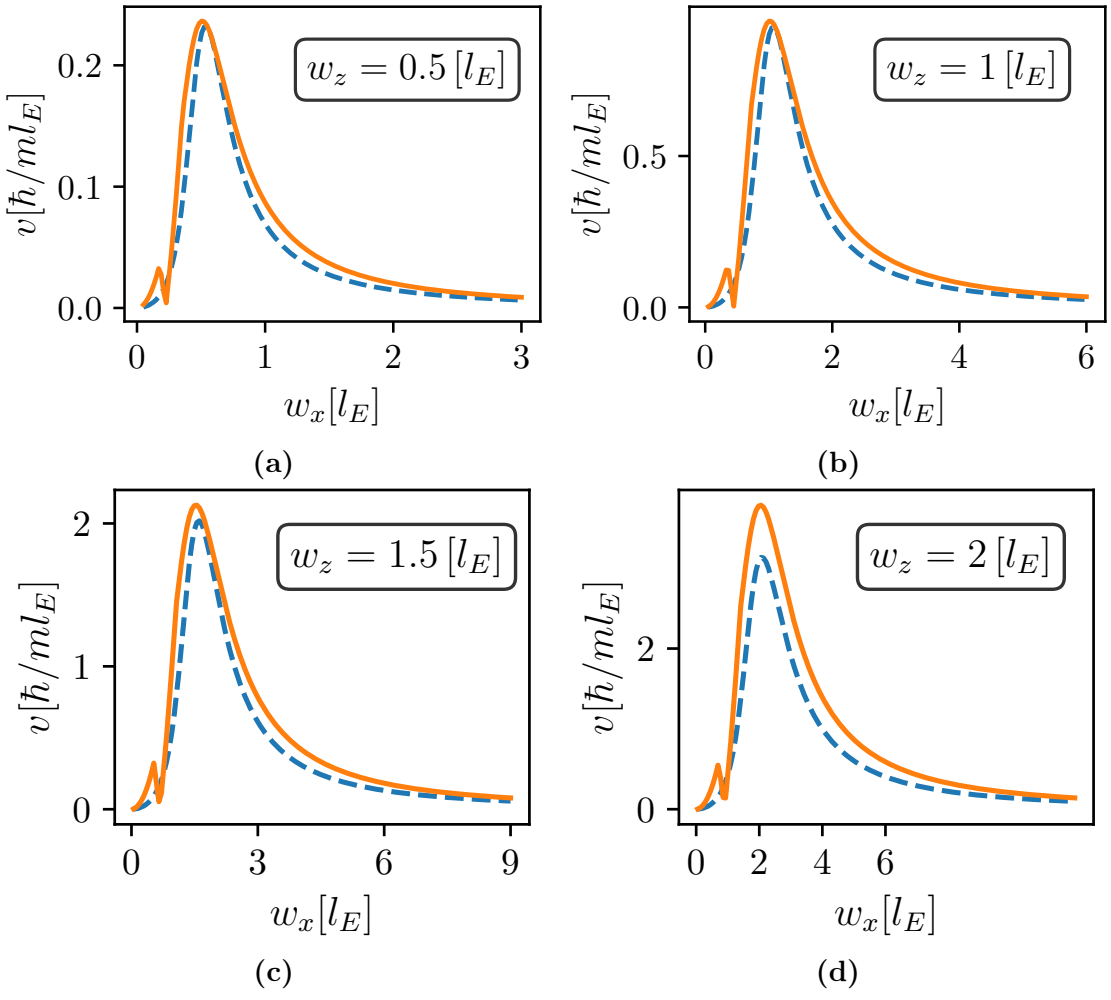


Figure 4.2.1: Comparison between analytical (orange lines) and numerical (blue dashed lines) values for the SOV as a function of w_x and for varying values of w_z. The values of v are given in units of $\hbar/ml_E$, and the values of w_x and w_z are given in units of l_E.

The fact that the analytical expression is more accurate for $w_z \lesssim l_E$ makes sense, as we expect the approximation to work best when the excited states and the

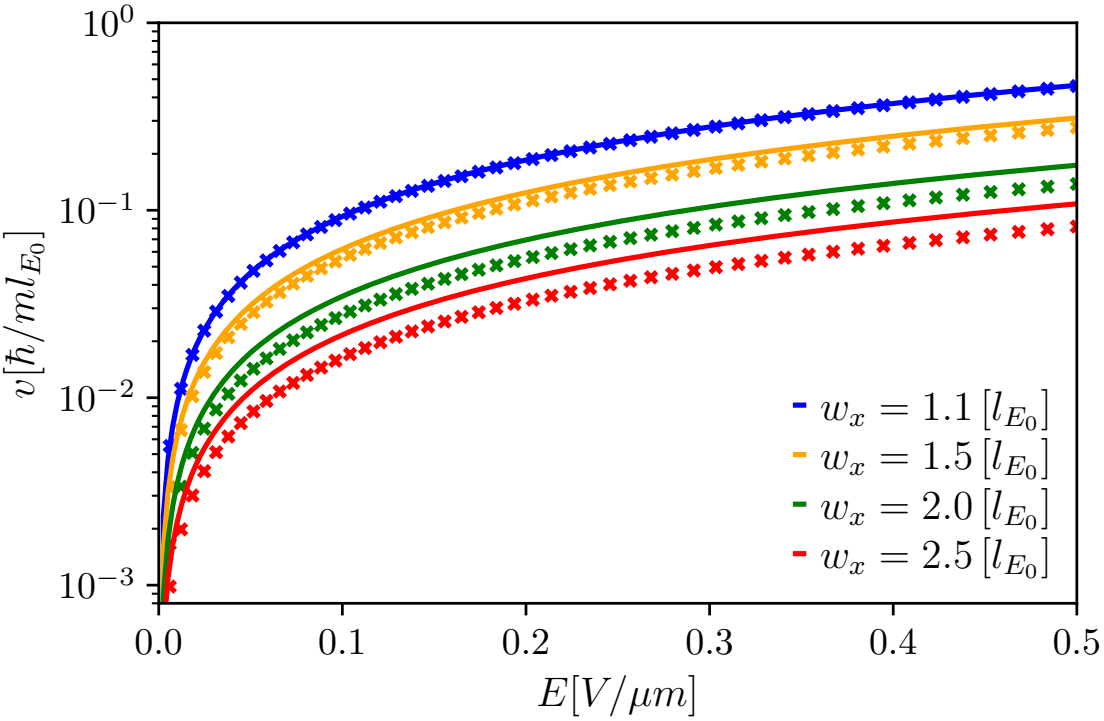


Figure 4.2.2: Comparison between numerical (crosses) and analytical (lines) values for the SOV v as a function of the electric field E for varying values of w_x and with $w_z = 1l_{E_0}$, where $E_0 = 1\text{V}\,\mu\text{m}^{-1}$.

ground states are split in energy by a large amount, which is exactly what the levels for a particle in a hard-wall potential do as the width of the potential thickness shrinks. The approximation of the wavefunctions of the LKBP Hamiltonian by the hard wall wavefunctions we also expect to work best when the electric field E is weak, which a large l_E signifies. This can be seen from Figure 4.2.2. In the plot, we show numerical and analytical values for the SOV as a function of the electric field E for varying values of w_x and a fixed value $w_z = 1$. One observes from the plot that the analytical values are indeed closer to the numerical ones for small E. We also observe that for $w_x = 1.1$, the analytical values match the numerical values to a high degree of accuracy. With increasing w_x, however, the values differ more, which is what we also observe in Figure 4.2.1.

The tradeoff for decreasing w_z or E is the strength of the SOV. We see, however, in Figure 4.2.1 that setting the width w_z to any less than 1 does not visibly increase the accuracy of our analytic expression. We thus find that we achieve the best possible compromise between the model's accuracy and the strength of the SOC by setting $w_z \sim l_E$.

We may further investigate the impact of introducing strain into our system by varying ε_0. We also compare the analytical results with numerical calculations to verify the accuracy of our model. We present in Figure 4.2.3 plots showing values of the SOV v for $w_z = 1$ and as a function of w_x for some fixed values of ε_0. From these comparisons, we observe that our analytical expression for v increases in magnitude as strain increases, as the numerical plots indicate. We note, however,

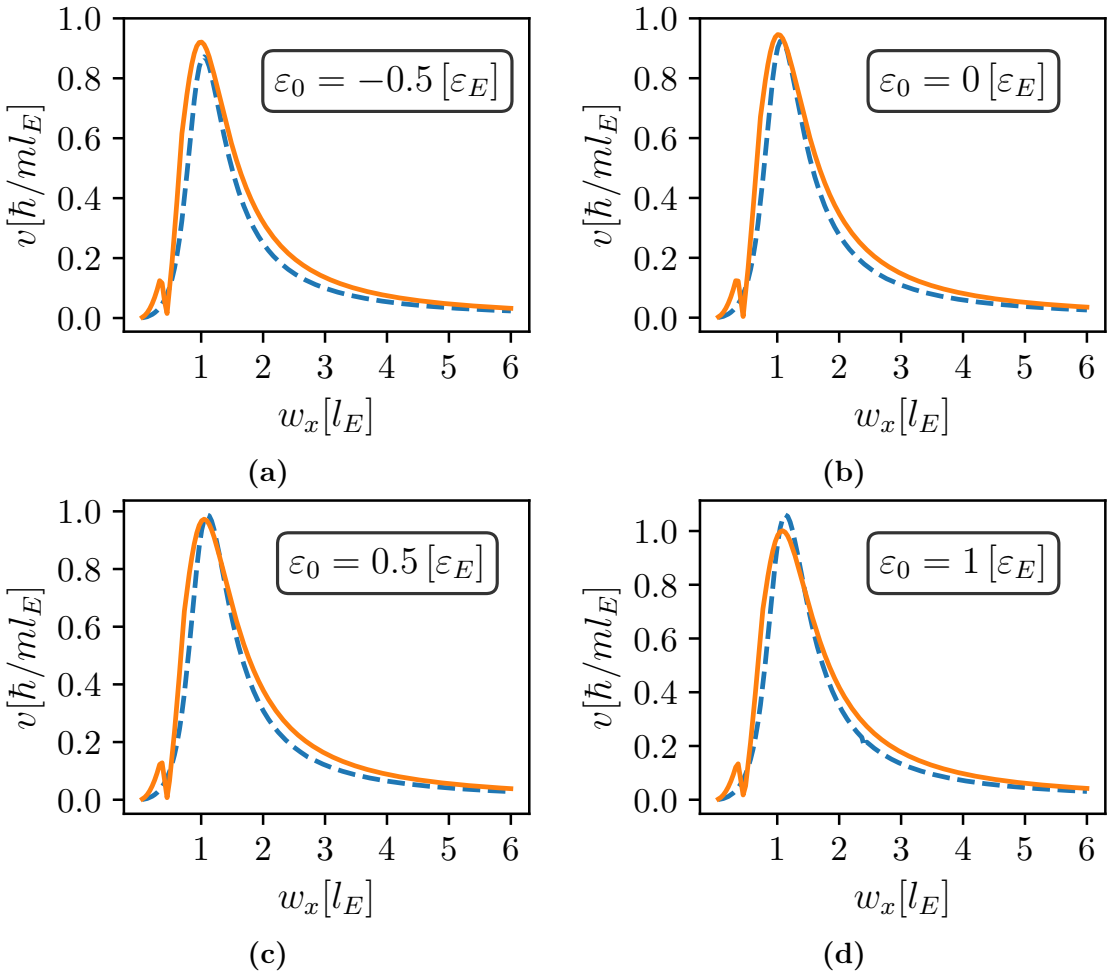


Figure 4.2.3: Comparison between numerical (blue dashed lines) and analytical (orange lines) values for the SOV as a function of w_x for varying values of ε_0, which is given in units of $\varepsilon_E = \hbar^2\gamma_1/2m|b|l_E^2$. In all the plots, we have fixed $w_z = 1$.

that the analytical values do not reproduce the dependence of v on the strain to a quantitatively accurate degree. We believe that one would have to work with a larger truncated Hilbert space for the analytical SOV to accurately capture the dependence on ε_0.

CHAPTER

FIVE

DISCUSSION

5.1 Model Validity

Our model of choice for investigating the SOC of a trapped hole in Ge has been the LKBP Hamiltonian for the two uppermost valence bands. In choosing this model over, say, other Kane models for a semiconductor, one implicitly assumes that the influence of the other bands present in the material can be disregarded. For this to be a valid assumption and approximation, the SO split-off energy Δ_{SO} needs to be large relative to the interaction between the topmost bands and the SO split-off band. If it is not, one would need to use the 6×6 LKBP Hamiltonian which includes the SO split-off band. One also assumes implicitly that the conduction band and the topmost valence band have a sufficiently large energy splitting.

We have also assumed the spherical approximation for the material parameters of Ge, ie. that $\gamma_2 = \gamma_3 \equiv \gamma_s$. This approximation assumes that γ_2 and γ_3 are similar enough such that cubic terms, which are proportional to $(\gamma_3 - \gamma_2)$, are small enough to neglect [41, 4]. In [24], one of the papers we have based much of our work on, the authors have used $\gamma_s = (\gamma_2 + \gamma_3)/2$. In [41], however, Baldereschi and Lipari find that a suitable value for the symmetric parameter that isolates isotropic and cubic parts of the LK Hamiltonian is $\gamma_s = (2\gamma_2 + 3\gamma_3)/5$. In our plots, we have employed the symmetric mean between γ_2 and γ_3, as Bosco et al. did. Still, in the literature, the more conventional choice of parameter appears to be that given by Baldereschi and Lipari. The choice to use $(\gamma_2 + \gamma_3)/2$ as our spherical parameter was made for a comparison between our results in this thesis and the results of Bosco et al. in [24] not to be obscured.

5.2 Validity and Error of Numerical Calculations

Our numerical calculations are based on discretizing the system's Hamiltonian, with our coordinate system a lattice. For this approximation to be representative of the continuous case, the scales of the system we are diagonalizing must be significantly larger than the relevant discretized scales. The relevant scales we must manage are length and energy scales.

5.2.1 Conditions for Length Scales

As mentioned in section 3.2, we discretize the coordinate system of the Hamiltonian to consist of a lattice of equally spaced points in each direction, such that the distance between neighboring points in the x-direction is a_x and a_z for neighboring points in the z-direction. For our numerics to yield accurate results, we must ensure that any physical lengths in our system are much larger than the relevant lattice point spacing. We then get 4 conditions:

$$w_x \gg a_x, \tag{5.1a}$$

$$w_z \gg a_z, \tag{5.1b}$$

$$L, l_E \gg a_x, a_z, \tag{5.1c}$$

$$l_x \gg a_x. \tag{5.1d}$$

Note that l_x is only relevant when we are imposing a harmonic potential in the x-direction, since it is a characteristic length scale of the harmonic potential. In our calculations, all lengths are in units of the arbitrary length L, which we, for the most part, set to $L = l_E$, including the lattice spacings. Since we have that

$$a_\alpha = \frac{w_\alpha}{N_\alpha + 1}, \tag{5.2}$$

we may ensure that all confinement widths are large enough by choosing large N's compared to the width of the system, meaning that we include enough lattice points in each direction. If we impose a tolerance δ_l for how much larger each length must be compared to each lattice spacing, we find that

$$N_\alpha > \frac{1}{\delta_l} - 1, \tag{5.3}$$

ensures that conditions in Equation 5.1 are satisfied. In the case of harmonic potential, it is also important that the width of the wavefunction l_x is small compared to the size of the full heterostructure, which in effect means that we must have that $l_x \lesssim w_x$. Employing the same tolerance as before, we get the condition that

$$\frac{a_x}{\delta_l} < l_x < (N_x + 1) a_x \delta_l. \tag{5.4}$$

In our numerical calculations we have used values of $N_\alpha \geq 50$ which obeys a tolerance of $\delta_l < 0.020$.

5.2.2 Conditions for Energy Scales

In the case of energy we must make sure that the energy scales of the systems are considerably larger than the energy associated with hopping between lattice sites $t_\alpha = \hbar^2/2ma_\alpha^2$ and that, for the case of a harmonic potential, that the energies of the system are much smaller than the potential energy of the confining potential at the boundaries of the lattice. The latter condition needs to be met so that the results of our calculations are minimally influenced by the hard wall potential which is ever present in a finite lattice. For a general energy quantity of our system E we must thus have that

$$E \ll \frac{\hbar^2}{2m \min(a_x, a_z)^2}. \tag{5.5}$$

This criterion is equivalent to the criteria of the length scales and is thus automatically fulfilled if the considerations in subsection 5.2.1 are fulfilled.

5.3 Validity and Accuracy of Analytical Results

As with every instance of producing approximate results, one must ensure that it is indeed justifiable to perform such an approximation. In our derivation, two fundamental approximations are being made: we are employing perturbation theory, and we are truncating our Hilbert space to include only a small finite subset of basis states. Both approximations need to be justified for our results to be valid.

The critical assumption being made in perturbation theory is that the perturbation is small compared to the unperturbed Hamiltonian. More precisely, interactions between basis states introduced by the perturbation should all be much smaller in magnitude than the energy splitting of the two eigenstates they couple. To second order the only contribution to the SOV between the two lowest energy states is mediated by a virtual state which couples to one of the HHs by a coupling which is $\propto E$ and to the other HHs by a coupling which is $\propto p_y/w_z$. Since we may let E and p_y be arbitrarily small, parameter ranges where perturbation theory is justifiable to second order exist. Problems arise however when going to third order or higher in perturbation theory. Then the SOC may be mediated via two or more virtual states. This could be a problem because some of the states in the $|n_x, n_z, s\rangle$-basis can lie very close in energy, like for example $|1, 2, 1\rangle$ and $|2, 1, 2\rangle$. These two states, and others, are coupled by a term proportional to $1/w_x w_z$, which cannot be changed without altering the system's eigenenergies. Such couplings thus risk invalidating the use of perturbation theory. The most critical shortcoming of our derivation seems thus to be the choice of basis for the LKBP Hamiltonian: the PIAB basis. One proposition to improve the approach we have presented in this thesis would thus be to work in a basis in which coupled states that are close in energy to the basis used in the present thesis are instead replaced by linear combinations of the states. For example one would substitute the PIAB basis states $|1, 2\rangle$ and $|2, 1\rangle$ for new states $\frac{1}{\sqrt{2}}(|1, 2\rangle - |2, 1\rangle)$ and $\frac{1}{\sqrt{2}}(|1, 2\rangle + |2, 1\rangle)$.

Another proposition for what one could do to fix the perturbation theory would

be to define the zeroth order Hamiltonian for our perturbation theory as

$$\begin{aligned}\underline{H_0}' =& \left(1+\frac{5\gamma_s}{2\gamma_1}\right)(\underline{p_{xx}}+\underline{p_{zz}})\otimes \mathbb{1}_{4\times 4} + \varepsilon_0 \mathbb{1}\otimes J_z^2 \\ &- \frac{2\gamma_s}{\gamma_1}\big(\underline{p_{xx}}\otimes J_x^2 + \underline{p_{zz}}\otimes J_z^2 + p_{xz}\otimes\{J_x,J_z\}\big),\end{aligned} \tag{5.6}$$

as opposed to treating the term $-(2\gamma_s/\gamma_1)p_{xz}\otimes\{J_x,J_z\}$ perturbatively, as we have done in the present thesis. This approach would eliminate the previously mentioned problem, since we would no longer need to treat the couplings $\propto 1/w_x w_z$ perturbatively. However, this approach is difficult to implement in practice because the eigenvectors become increasingly difficult to compute analytically. Including only the four PIAB basis states $|1,1\rangle$, $|1,2\rangle$, $|2,1\rangle$, and $|2,2\rangle$ makes $\underline{H_0}'$ a 16×16 matrix where it needs to be diagonalized in 4×4 blocks.

Lastly, one could derive the SOV using the eigenfunctions of the LK Hamiltonian, thereby omitting the PIAB approach entirely. Wave functions are given in [31], for example.

5.4 Future Work

To supplement our work here, we propose some future work on the problem. This includes investigating materials other than Ge, exploring alternative semiconductor models, and applying the theory to quantum technology applications.

In our model, we used the so-called spherical approximation for the LK Hamiltonian. The assumption is that material parameters γ_2 and γ_3 are close enough that we may set them both equal to a spherical parameter γ_s. This is justified in Ge, but not for all zinc blende semiconductors. A natural extension to the present work would thus be to consider the LK Hamiltonian without the spherical approximation. In this way one could derive explicit expressions for the SOC that can describe a larger fraction of materials.

As we can see from some of the plots in [24], the growth direction of the Ge can have a large impact on its properties. It would thus be interesting to investigate how the system's growth direction affects the system's SOC. A procedure for doing so is described in [25]. The general approach would be to work in a rotated coordinate system, in which the parameters depend on the rotation.

We have taken great inspiration from [24] in this thesis, as we have been interested in building upon their proposal for a qubit by deriving some analytic expressions for the SOC and further to derive analytic expressions for the magnetic field and effective mass of the effective Hamiltonian. Using this one could either try to reproduce the results of Bosco et al and to try something new with the system at hand. One could for example try to control the qubit by tuning the width in the x-direction rather than translating the confinement using an alternating current as they do in their paper. The hope would then be to develop a qubit proposal

that incorporates the strengths of previous hole spin qubits while providing a more effective control handle than those previously proposed.

Lastly, we note that the study of SOC in semiconductors is of great importance also in the field of hybrid superconducting systems. Understanding how the SOC alters the properties of such systems will be critical to extracting the full potential in future applications. We highlight a paper by Vezzosi et al [26] where they look at an InP/GaSb hybrid device exhibiting strong SOC. Similarly to what we have done here, they derive an effective Hamiltonian along the wire, as well as a SOC parameter. A similar approach to their method could be made for our problem or vice versa. This highlights the relevance of our method or study to the broader spectrum of applications in condensed matter physics.

CHAPTER

SIX

CONCLUSIONS

In this thesis, we have investigated the spin-orbit coupling of the lowest-energy heavy-hole subspace in a uniaxially strained germanium heterostructure with an external constant electric field applied along the strain axis. We have derived an explicit expression for the strength of the spin-orbit coupling, parametrized by what we refer to as the spin-orbit velocity v, using a second-order Schrieffer-Wolff transformation in the limit where the external electric field is weak. To simplify the transformation, we have rewritten the xz-part of the 4×4 Hamiltonian, thus neglecting the effect of the SO split-off band, Luttinger-Kohn-Bir-Pikus Hamiltonian with spherical material parameters in terms of the 2-dimensional particle-in-a-box eigenbasis and truncated the full Hilbert space to include only 12 basis states. We then extracted the spin-orbit velocity by tracing the effective Hamiltonian for the lowest-energy heavy-hole subspace using the Pauli spin matrices. The resulting expression is given in Equation 4.1.

To investigate the accuracy of our expression, we have compared it to numerical values for the spin-orbit velocity, which we have calculated by discretizing the Luttinger-Kohn-Bir-Pikus Hamiltonian, numerically diagonalizing it, and then calculating v using first-order perturbation theory. From this analysis, we find that our approach is effective in qualitatively capturing the dependence of the spin-orbit velocity when the electric field is weak, and that the confinement in the direction of the electric field, w_z, is thin. In this parameter regime and when the confinement width in the other confined direction w_x is similar to w_z, our expression also captures quite well the magnitude of the spin-orbit velocity. The expression we have derived also qualitatively captures the dependence of the spin-orbit velocity on the system's strain, namely that, for a small strain energy $|b|\varepsilon_0$, the magnitude of the spin-orbit velocity increases with strain. Our analytical expression, however, does not quite capture this dependence quantitatively.

To second order, we may identify parameter regimes where the perturbative approach is justified, but we find that to perform higher-order perturbative approximations, one must handle large couplings between states that are close in energy. We believe one could omit such large coupling terms invalidating the use of perturbation theory by working in a more suitable basis than the two-dimensional particle-in-a-box-basis.

Despite the limitations of the results obtained in this thesis, we have provided a compact analytical expression that qualitatively describes the behavior of the SOC for low-energy heavy holes in the topmost valence band of strained germanium. The method we have outlined, adapted from [25], is a general analytical guide for deriving such an expression, and we hope that further work may improve upon the work presented in this thesis.

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APPENDICES

A - DIAGONALIZING A 2×2 REAL HERMITIAN MATRIX

We derive in this appendix a rotational matrix R that diagonalizes a real Hermitian matrix

$$H = \begin{pmatrix} a & d \\ d & b \end{pmatrix}, \tag{1}$$

where we assume $a, b, d \in \mathbb{R}$. We define the rotational matrix

$$R \equiv \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}. \tag{2}$$

It can be shown that d must be real for R to diagonalize H. We also note that a Hermitian matrix may always be diagonalized using a unitary transformation. It is easy to show that R is indeed unitary and that $R^\dagger = R^T$. We want to solve the equation

$$R^T H R = \begin{pmatrix} ac^2 + bs^2 + 2dsc & (b-a)sc + d\cos 2\phi \\ (b-a)sc + d\cos 2\phi & as^2 + bc^2 - 2dsc \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}, \tag{3}$$

where we have used $s \equiv \sin\phi$, $c \equiv \cos\phi$, and trigonometric double angle identities. We solve for ϕ in the off-diagonal equations and get that

$$\phi = \frac{1}{2}\arctan\left(\frac{2d}{a-b}\right). \tag{4}$$

Inserting Equation 4 into $\sin\phi$ and $\cos\phi$ and writing the expressions in exponential form, we may simplify them and get that

$$\begin{aligned} \cos\phi &= \frac{1}{\sqrt{2}}\sqrt{1 + \frac{|a-b|}{\sqrt{(a-b)^2 + 4d^2}}}, \\ \sin\phi &= \frac{1}{\sqrt{2}}\sqrt{1 - \frac{|a-b|}{\sqrt{(a-b)^2 + 4d^2}}}. \end{aligned} \tag{5}$$